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EAS 6134: HW 6
Nonlinear Inverse Methods

1. The Arrhenius Relationship: parameterizing the thermally-activated semiconduction σ:

$$\sigma(T) = \sigma_0 e^{-A/k_B T} \quad (1)$$

as a function of temperature T. The parameters are σ_0 (conductivity at ∞ K), the limiting conductivity, and A, the activation energy.

a) The data provided is depicted in the Figure following this page.

b) Parameterized Least Squares can readily be carried out with a ~~pa~~ simple transform of the data:

$$\sigma(T) = \sigma_0 e^{-A/k_B T}$$

$$\Rightarrow \log \sigma(T) = \log \sigma_0 e^{-A/k_B T}$$

$$\Rightarrow \log \sigma(T) = \log \sigma_0 + \log e^{-A/k_B T}$$

let $s \equiv \log \sigma_0$ $\quad \Rightarrow \log \sigma(T) = s + -\frac{A}{k_B T}$

let $x \equiv \frac{1}{k_B T}$ $\quad \Rightarrow \log \sigma(T) \rightarrow \log \sigma(x) = s - Ax, \quad (2)$

where σ_0 is e^s and A remains. By fitting this transformed & now linear expression using Least-Squares, we can obtain estimates for the model parameters A, σ_0. Because the data errors are not known specific to each point, we do not employ a weighted approach. Since the material in question dominates the upper mantle (Dunite), we assume (initially) that A, σ_0 are the only parameters, and are constant across temperature.

However, if we work in SI units (i.e., Kelvin / Joules), then the small value of Boltzmann's Constant causes a high condition number in $\underline{\underline{G}}$. That is, from eqn (2), we have

$$\vec{m} = \begin{pmatrix} s \\ A \end{pmatrix} = \begin{pmatrix} \log \sigma_0 \\ A \end{pmatrix} \; ; \; \vec{d} = \begin{pmatrix} \log(\sigma(T_1)) \\ \log(\sigma(T_2)) \\ \vdots \end{pmatrix}, \text{ and hence}$$

$$\underline{\underline{G}} = \begin{pmatrix} 1 & -x_1 \\ 1 & -x_2 \\ \vdots & \vdots \\ 1 & -x_n \end{pmatrix} \quad \text{where } x = \frac{1}{k_B T}.$$

With $T \sim \mathcal{O}(1000K)$ and k_B in Joules, the second column of $\underline{\underline{G}}$ is $\mathcal{O}(-10^{19})$. We apply a form of pre-conditioning $\underline{\underline{G}}$ by converting to energy dimensions of eV: this yields a condition number of $\text{cond}(\underline{\underline{G}}_{eV}) = 87$ compared to a diverging value for $\text{cond}(\underline{\underline{G}}_J) \sim \infty$.

Also, to retain the dimensionless requirement for the argument of log( ), we take $d_i = \log\left(\sigma(T_i) / (1\ S/m)\right)$

c) The above procedure yields the following values:

$$\text{Model} \begin{cases} \sigma_0 = 65.131 \ S/m \\ A = 1.5879 \ eV \end{cases}$$

The fit is fairly strong compared against the data, both when in the transformed format and as described by the original equation (1).

d) The error of 5% maps into the errors of our expected estimates, or uncertainties, in a non linear way. First, $e^{d_i} = \sigma_{obs}(T)$, or $\vec{d} = \begin{pmatrix} \log(\sigma_{obs}(T_i)) \\ \vdots \end{pmatrix}$. However, notice that for gaussian noise with a value of 5%,

~~log(x+ε) ≈~~

$$\log \sigma_{obs}(T_i) = \log(\sigma_{true}(T_i) + \varepsilon),$$

so the error is no longer normally-distributed. This means that our approach will favor error at lower temperature, since the log of the error distribution is no longer normal.

cont.)

We compute the covariance matrix $\underline{\underline{C}}_m$ to quantify the variance, or uncertainty, in each parameter:

$$\underline{\underline{C}}_m = \begin{pmatrix} 3.82 \cdot 10^{-3} & 4.01 \cdot 10^{-4} \\ 4.01 \cdot 10^{-4} & 4.26 \cdot 10^{-5} \end{pmatrix}, \qquad (3)$$

indicative of fairly small uncertainties in $\log \sigma_0$ (top) and A (bottom). To view this as a direct uncertainty on σ_0, we calculate:

$$C_{11} \equiv A_s \sim \log \sigma_0 \Rightarrow e^{A_s} \sim e^{\log \sigma_0} = \sigma_0$$

$$\Rightarrow \text{uncertainty on } \sigma_0 \text{ is } \approx 1.001,$$

which represent $\frac{1.001}{65.1312} \approx 1.5\%$ uncertainty.

The correlation between the fit and the true data is clear in the residuals (see below Figure). The residuals are shown alone, and with a (-)5% error curve for comparison, while the residuals remain below the 5% error threshold, they are behaved in a highly non-Gaussian way. This may stem from using an error threshold slightly too high as well as the fact that the fit was transformed, as discussed above. The correlation coefficient obtained from $\underline{\underline{C}}_m$ is cor ~ 0.99, indicating that we did not meet the ideal parameterization. Also, the highly non-random nature of the residuals (i.e., the apparent 'oscillation') indicates that our model may not be of the ideal mathematical form to precisely fit the data.

e) To recompute the analysis with the entire dataset, we modify our model: in obtaining the Arrhenius relationship for the temperature dependence of conductivity, a factor of $1/T$ is often 'absorbed' into the constant σ_0. Hence, we test a relationship $\sigma(T)$ of the form

$$\sigma(T) = \sigma_0 e^{-A_0/k_B T} + \frac{\sigma_1}{T} e^{-A_1/k_B T}$$

[e) cont.] This relationship captures the full 'expected' physical behavior for the (semi) conductivity of Dunite as a function of ambient temperature. This additional term makes our approach a highly ill-posed, nonlinear optimization problem. However, we now account for both small polaron hopping at lower temperatures ~~and~~ (i.e., with the $\frac{1}{T}$ pre-factor) and for ionic vacancy transport, which is described by the 'original' model.

To perform this inversion, we therefore use the Levenberg-Marquardt algorithm. The uncertainty for each point is given by some fraction Ω of its value, between 1-5% (see part (d)). We use the Chi-squared objective function statistic:

$$\vec{M} = \begin{pmatrix} \sigma_0 \\ \sigma_1 \\ A_0 \\ A_1 \end{pmatrix} \quad ; \quad \chi^2(\vec{M}) = \sum_{i=1}^{N} \left(\frac{\overset{\text{data}}{\sigma_i} - \overset{\text{model output}}{\sigma(T_i,\vec{M})}}{(\Omega \sigma_i)} \right)^2$$

The weighted residuals are $r_i = \left(\frac{\sigma_i - \sigma(T_i,\vec{M})}{\Omega \sigma_i} \right)$. The Objective function for Levenberg-Marquardt is a function of the Jacobian J_{ij}, written as

$$J_{ij} = \frac{\partial r_i}{\partial M_j}$$

It uses a parameter λ that acts mainly as a balance between Gauss-Newton and Gradient descent.

The results are included in the following figures. We show the model output alone, and compared against the 'transformed' LS fit to the truncated dataset. Likely due to our choice of model, the truncated fit with the simpler model works better. We also depict the residuals, the behavior of which indicate that there is a systematic (curve-like) non-Gaussian misfit. The results use an error threshold of $\Omega = 1\%$, and do not change strongly in the range $\Omega \in [1\%, 5\%]$.

F) To do so, we would apply a Markov chain Monte Carlo method that also includes parameters representing the weights of different model behaviors, i.e., the terms of $\sigma(T,\vec{M})$. This allows us to determine the posterior distribution as a function of the model itself, and would allow for the optimization of our model / objective function 'on the fly'.